

GRAIN-CAN: Preserving Local CAN Behavior Changes for Cross-Vehicle Intrusion Detection

Qi’ao Li

School of Cyber Science and Technology, Beihang University
qi_aolee@buaa.edu.cn

Abstract. Cross-vehicle and unknown-attack CAN intrusion detection is difficult because vehicle-specific CAN distributions and attack scripts change across train and test settings. Raw fixed-window representations can also dilute short local attack evidence. We propose GRAIN-CAN, a supervised feature-based CAN IDS pipeline that extracts same-ID timing gaps, payload changes, payload statistics, and local CAN-ID behavior before fixed-window aggregation. These features describe relative local behavior changes rather than only absolute CAN patterns of one vehicle. The aggregated features are then given to a supervised classifier that outputs a normal/attack decision and, when supported, an attack score. We evaluate CT&T test01–test04 with conventional IDS metrics, confusion matrices, and controlled baselines, including all-normal, reproduced public-style, raw-window, rule/statistical, and safe-feature rows where available. AUPR and fixed-FPR detection rates are reported as supplementary score-based metrics. GRAIN-CAN improves the corrected baseline under several shifted settings, but it does not guarantee arbitrary unseen-attack detection.

Keywords: CAN bus · Intrusion detection · Automotive security · Feature-based learning · Cross-vehicle detection

1 Introduction

CAN IDS observes only timestamps, arbitration IDs, DLCs, and payload bytes. The need for such detectors follows from long-standing demonstrations that in-vehicle networks expose practical attack surfaces and that CAN lacks built-in message authentication [10, 3, 13]. In cross-vehicle unknown-attack evaluation, both the vehicle and the attack family may change between training and testing. A detector that mainly memorizes absolute CAN-ID frequencies, payload ranges, or attack-script IDs can fail when these distributions move. A long raw window can also bury a short injection or spoofing artifact under many normal frames.

GRAIN-CAN addresses this by extracting local behavior changes before aggregation. For each frame, it compares the current message with recent history of the same CAN ID and records timing gaps, payload differences, payload statistics, and local ID behavior. This design is close in spirit to timing/statistical IDS lines [17, 18], but it keeps multiple local behavior signals together before

fixed-window aggregation. A fixed window then aggregates these explicit local signals, and a supervised classifier learns the normal/attack boundary. GRAIN-CAN reduces, but does not eliminate, dependence on vehicle-specific absolute CAN distributions. It does not recognize unseen attack names; it detects local behavior disruptions that may be shared by injection, spoofing, replay, flooding, and masquerade attacks. It may miss attacks that preserve timing, payload continuity, and local CAN-ID behavior.

Our contributions are limited to two points. First, we propose GRAIN-CAN, a supervised feature-based CAN IDS pipeline that converts raw frames into same-ID local behavior changes before window aggregation. Second, we evaluate it under CT&T cross-vehicle and unknown-attack settings with conventional IDS metrics and explicit confusion matrices, comparing against all-normal, public-style classical, raw fixed-window, rule/statistical, and safe-feature baselines where available.

2 Background and Motivation

2.1 CAN IDS and Cross-Shift

Classical CAN provides compact message fields but no semantic signal names. IDS methods infer attacks from timing, payload evolution, ID frequencies, transition structure, learned sequence models, or physical-layer artifacts [4, 5, 17, 18]. Recent learning-based work has explored BERT-style sequence modeling, signal-level deep learning, statistical graph learning, federated multi-stage detection, and state-space sequence models [1, 15, 20, 2, 12]. CT&T and ROAD have made shifted CAN IDS easier to study, but they also show that train/test protocol and metric definitions matter [19, 11, 7, 9]. Cross-vehicle unknown-attack detection is harder because training and testing differ in both vehicle behavior and attack behavior.

2.2 Relation to Recent Work

Recent CAN IDS work includes fingerprinting, statistical timing models, sequence encoders, graph-style representations, and benchmark studies [16, 7, 8]. These lines are complementary to GRAIN-CAN: we do not introduce a new deep backbone or physical-layer fingerprint, and we do not claim to replace strong recent models such as signal-level autoencoders, graph IDS, or Mamba-style sequence baselines [15, 20, 12]. Instead, we ask whether causal same-ID local behavior features, kept before window aggregation, improve a supervised detector under cross-shift evaluation. Because rare-attack settings can make accuracy-like metrics misleading, we report attack-positive precision, recall, F1, FPR, FNR, and confusion matrices first, with AUPR and ROC-style operating metrics as supplementary evidence [6, 14].

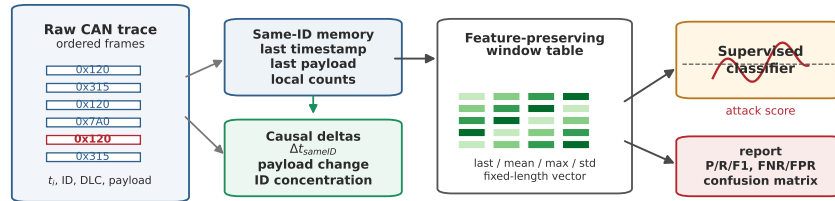
2.3 Local Evidence Dilution

Many CAN attacks affect only a small number of frames or a small subset of IDs. If a model receives a raw long window, the abnormal frames may be numerically diluted by normal traffic. GRAIN-CAN makes the local change explicit first, then aggregates it. This design choice is intentionally modest: it does not assume that one statistic can describe all attacks, and it does not rely on a new deep backbone. It asks whether the evidence already present in CAN timing and payload continuity can be kept visible long enough for a supervised detector to use it.

a GRAIN-CAN feature-preserving CAN IDS

Causal same-ID local behavior is extracted before fixed-window aggregation and supervised detection.

b Mechanism: preserve local evidence before aggregation



c Fixed evaluation contract



Fig. 1. GRAIN-CAN pipeline. Same-ID local history is converted into timing, payload-change, and ID-behavior features before fixed-window aggregation and supervised classification.



Fig. 2. Motivation for feature preservation. Raw long windows can hide a short burst of abnormal frames, whereas GRAIN-CAN exposes local same-ID changes before aggregation.

3 GRAIN-CAN Method

3.1 Overview

GRAIN-CAN is a supervised feature-based CAN intrusion detection pipeline, not a neural network architecture. It converts raw CAN frames into same-ID local behavior changes before window aggregation and uses a supervised classifier to learn a normal/attack decision boundary.

3.2 Input, Output, and History Buffer

The input is an ordered stream $x_i = (t_i, id_i, dlc_i, b_{i,1}, \dots, b_{i,8})$. Labels are used only for supervised training and evaluation. During feature extraction, GRAIN-CAN maintains a per-ID history buffer with the last timestamp, last payload, and local ID statistics. The output is a binary normal/attack decision and, when the classifier supports it, an attack score.

3.3 Local Behavior Features

For a frame with CAN ID id_i , the method computes the same-ID time gap, payload difference from the previous same-ID frame, payload summary statistics, and local CAN-ID behavior. These are relative local changes rather than raw absolute distributions alone.

The main feature groups are: (i) same-ID timing, including the gap between the current frame and the previous frame with the same CAN ID; (ii) payload dynamics, including byte values and the L_1 change from the previous payload of the same ID; (iii) payload statistics, such as sum, mean, and standard deviation; and (iv) local ID behavior, including ID frequency and short-window concentration. All history-dependent quantities are causal. The extractor never uses a future frame, a test label, or a statistic fitted on the test split.

3.4 Fixed Window Aggregation and Classifier

Frame-level features are aggregated using a fixed pre-test window length, such as W10, W20, or W100. The window label is attack if any frame in the window is labelled attack. The classifier is a supervised score-producing model; the current GRAIN rows use GradientBoosting-style tree classifiers in the available repository results.

For a window $W_j = \{x_s, \dots, x_e\}$, GRAIN-CAN applies simple summary operators over the frame-level features: last value, mean, standard deviation, minimum, maximum, and selected count/entropy features where available. These operators are deliberately transparent. They keep strong local changes, such as a large same-ID timing gap or payload delta, while still producing a fixed-length representation for conventional supervised classifiers.

Algorithm 1: GRAIN-CAN training and inference. Training initializes per-ID history states, processes training frames in temporal order, computes same-ID time gaps, payload differences, payload statistics, and local ID behavior, aggregates fixed windows, trains a supervised classifier with normal/attack labels, and selects a threshold on validation data if scores are available. Inference freezes feature definitions, classifier, window size, and threshold before processing test frames. It uses only current and previous frames; test labels, future frames, test-set statistics, test-set thresholds, and test-set window selection are not used.

3.5 Relation to Rule-Based and Statistical IDS

GRAIN-CAN uses interpretable timing, payload, and ID-behavior signals, so it is related to rule-based and statistical CAN IDS. The difference is that it does not alarm on a manually selected threshold over one signal. It uses these signals as supervised features and learns a joint decision boundary.

4 Experimental Evaluation

4.1 Datasets, Baselines, and Metrics

CT&T set01 is the main dataset. It provides four shifted settings: test01 (known vehicle, known attack), test02 (unknown vehicle, known attack), test03 (known vehicle, unknown attack), and test04 (unknown vehicle, unknown attack). ROAD and other data are used only as sanity checks where compatible results are available.

Baselines are grouped as all-normal, classical ML, raw-window neural, safe-feature, rule/statistical, and GRAIN rows. We report conventional IDS metrics:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

where P is precision, R is recall or detection rate, TP is detected attacks, FP is false alarms, and FN is missed attacks. We also report

$$FNR = \frac{FN}{TP + FN}, \quad FPR = \frac{FP}{FP + TN}.$$

where FNR measures missed attacks and FPR measures false alarms on benign traffic. AUPR, AUROC, and detection rate at fixed FPR are supplementary when comparable scores exist:

$$\text{TPR}(\tau) = \frac{TP(\tau)}{TP(\tau) + FN(\tau)} \quad \text{s.t.} \quad \text{FPR}(\tau) \leq \alpha.$$

where τ is the threshold and α is the false-alarm-rate budget.

Table 1. Dataset and class-balance summary for settings used in the revised evidence package.

Dataset	Usage	Frames	Windows	Attack ratio
road	main, few-label, ablation, efficiency	28244539	282192	0.0331
ctt_test01	main, few-label, generalization, ablation, low-FPR	16355810	163547	0.0069
ctt_test02	unknown vehicle, known attack	17101057	171000	0.0125
ctt_test03	known vehicle, unknown attack	19288415	192872	0.0036
ctt_test04	unknown vehicle, unknown attack	23873695	238725	0.0027
crysys_family_mod_subset	optional external sanity/generalization	38503318	384951	0.1712
hcr1_can_intrusion	sanity check	4613439	46126	0.4864
car_hacking	sanity check	17558346	175576	0.1328

4.2 Implementation and Selection Protocol

GRAIN-CAN is evaluated as a fixed supervised pipeline. Feature definitions are causal; classifier training uses the recorded training data; and the test split is not used to choose feature definitions, thresholds, or window lengths. Because the repository contains several historical result tables, we keep protocol-audit tables in the main paper rather than asking the reader to infer what was fixed and what was only sensitivity analysis.

Table 2. GRAIN-CAN classifier and feature protocol.

Setting	classifier	feature_groups	score_available	test_used_for_selection
ctt_test01-test04	GradientBoostingClassifier for GRAIN rows	same-ID timing[payload bytes payload statistics payload delta CAN-ID behavior]	True	False

4.3 Main Results

The corrected CT&T results show that GRAIN rows are competitive with reproduced baselines under conventional attack-positive metrics. The results should

Table 3. Window-length selection audit. Window rows are sensitivity evidence, not test-time selection.

Setting	candidate	validation	selected_by	selected_window	selected_by_test
ctt_test01	10 20 100			False none; report W10/W20/W100 separately	False
ctt_test02	10 20 100			False none; report W10/W20/W100 separately	False
ctt_test03	10 20 100			False none; report W10/W20/W100 separately	False
ctt_test04	10 20 100			False none; report W10/W20/W100 separately	False

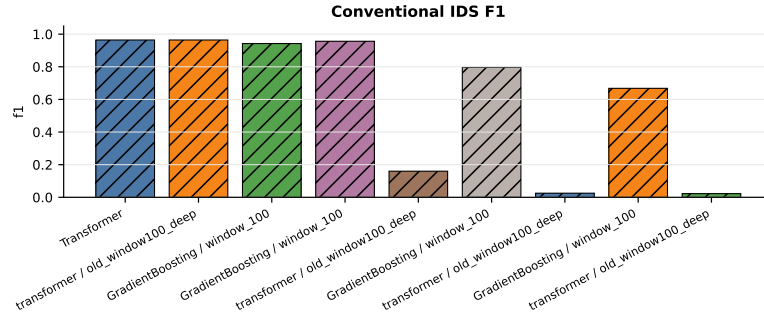
Table 4. Threshold-selection audit.

Setting	Model	score	available	threshold	selected_on	test_used_for_threshold
ctt_test04	GRAIN-W100				True validation or stored default; best-test rows supplementary only	False

Table 5. Conventional IDS metrics for representative CT&T rows. Attack is the positive class.

Setting	Model	P	R	F1	FNR	FPR
ctt_test01	GradientBoosting / sample	0.8600	0.9805	0.9163	0.0195	0.0011
ctt_test01	GradientBoosting / window_10	0.9782	0.9530	0.9654	0.0470	0.0001
ctt_test01	GradientBoosting / window_100	0.9990	0.8918	0.9424	0.1082	0.0000
ctt_test01	GradientBoosting / window_20	0.9944	0.9441	0.9686	0.0559	0.0000
ctt_test01	GradientBoosting / window_5	0.9585	0.7979	0.8708	0.2021	0.0002
ctt_test01	GradientBoosting / window_50	0.9990	0.9105	0.9527	0.0895	0.0000
ctt_test01	MLP / sample	0.2093	0.3652	0.2661	0.6348	0.0096
ctt_test01	Transformer	1.0000	0.9309	0.9642	0.0691	0.0000
ctt_test01	transformer / old_window100_deep	1.0000	0.9309	0.9642	0.0691	0.0000
ctt_test02	GradientBoosting / sample	0.9314	0.9981	0.9636	0.0019	0.0009

not be read as a universal unknown-attack solution: test04 remains the hardest setting, and recall/FNR show nontrivial misses. Table 5 deliberately reports precision, recall, F1, FNR, and FPR rather than only aggregate accuracy. A detector that reports high aggregate scores but misses the rare attack class is not sufficient for an IDS.

**Fig. 3.** Representative conventional F1 results. The chart uses reproduced or locally computed rows only.

4.4 Confusion-Matrix Evidence

Confusion matrices make the false-alarm and missed-attack tradeoff explicit. This is important because CT&T test04 is strongly normal-dominated; a method can look acceptable under aggregate metrics while still producing zero useful attack detection. We therefore keep TP, FP, TN, and FN as first-class reporting fields.

Table 6. Representative confusion-matrix counts. Attack is positive.

Setting	Model	tp	fp	tn	fn
ctt_test01	GradientBoosting / sample	1106.0000	180.0000	162239.0000	22.0000
ctt_test01	GradientBoosting / window_10	1075.0000	24.0000	162395.0000	53.0000
ctt_test01	GradientBoosting / window_100	1006.0000	1.0000	162418.0000	122.0000
ctt_test01	GradientBoosting / window_20	1065.0000	6.0000	162413.0000	63.0000
ctt_test01	GradientBoosting / window_5	900.0000	39.0000	162380.0000	228.0000
ctt_test01	GradientBoosting / window_50	1027.0000	1.0000	162418.0000	101.0000
ctt_test01	MLP / sample	412.0000	1556.0000	160863.0000	716.0000
ctt_test01	Transformer	1050.0000	0.0000	162419.0000	78.0000

4.5 Cross-Shift Decomposition

The four CT&T settings isolate different sources of shift. Test02 changes the vehicle but keeps attack families known; test03 keeps the vehicle known but changes the attack family; test04 changes both. This decomposition is why test04 should be interpreted as the central cross-vehicle unknown-attack stress case rather than as one more ordinary test split.

Table 7. CT&T shift decomposition with GRAIN window rows.

Setting	test_vehicles	test_attacks	Model	F1	AUPR
ctt_test01	known	known	GRAIN-W10	0.9655	0.9330
ctt_test01	known	known	GRAIN-W20	0.9685	0.9413
ctt_test01	known	known	GRAIN-W100	0.9423	0.8960
ctt_test02	unknown	known	GRAIN-W10	0.9948	0.9901
ctt_test02	unknown	known	GRAIN-W20	0.9927	0.9866
ctt_test02	unknown	known	GRAIN-W100	0.9564	0.9354
ctt_test03	known	unknown	GRAIN-W10	0.6765	0.6778
ctt_test03	known	unknown	GRAIN-W20	0.6677	0.7094

4.6 Raw Fixed-Window Versus GRAIN-CAN

This comparison tests whether local timing and payload-change evidence before window aggregation improves detection compared with raw fixed-window representations.

Table 8. Raw fixed-window and GRAIN-CAN comparison.

Setting	Model	W	P	R	F1	AUPR
ctt_test01	GRAIN-W100	100	0.9988	0.8919	0.9423	0.8960
ctt_test01	Old raw window100 Transformer	100	1.0000	0.9307	0.9641	0.9844
ctt_test02	GRAIN-W100	100	0.9836	0.9306	0.9564	0.9354
ctt_test02	Old raw window100 Transformer	100	0.0870	1.0000	0.1600	0.0489
ctt_test03	GRAIN-W100	100	0.9996	0.6623	0.7967	0.9250
ctt_test03	Old raw window100 Transformer	100	1.0000	0.0133	0.0263	0.2708
ctt_test04	GRAIN-W100	100	0.9365	0.5189	0.6678	0.7845
ctt_test04	Old raw window100 Transformer	100	0.0113	0.9794	0.0224	0.1733

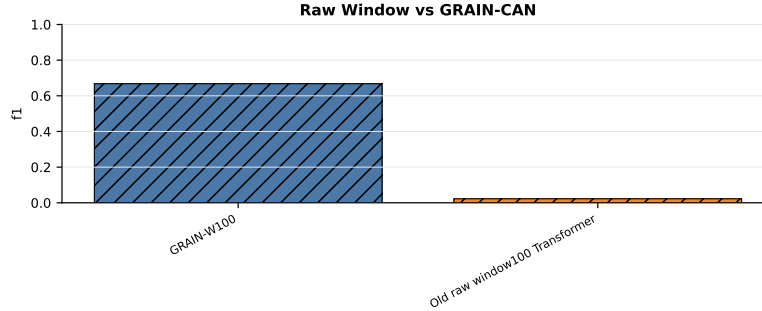


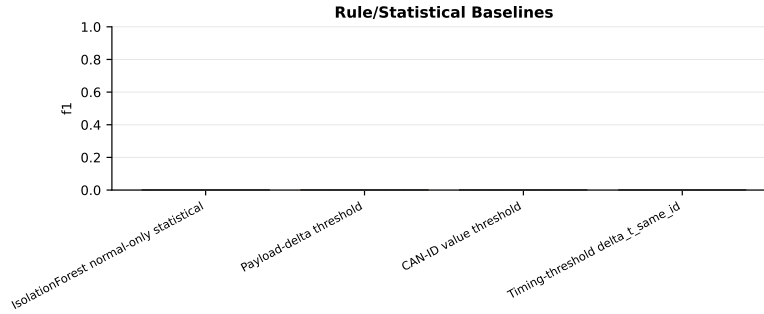
Fig. 4. Raw fixed-window representation versus GRAIN-CAN on CT&T test04. The result supports preserving local behavior changes before aggregation.

4.7 Rule and Statistical Baselines

Single-feature threshold rules and normal-only statistical anomaly detection test whether GRAIN-CAN is merely a rule detector. The results show that individual signals can be useful, especially timing, but the supervised pipeline is the method-level object studied here.

Table 9. Rule/statistical baselines on CT&T test04 with validation-selected thresholds.

Model	P	R	F1	FPR	FNR	AUPR
Timing-threshold delta_t_same_id	0.0009	0.8151	0.0018	0.9999	0.1849	0.0898
Payload-delta threshold	0.0011	1.0000	0.0022	1.0000	0.0000	0.0533
CAN-ID value threshold	0.0010	0.9266	0.0020	1.0000	0.0734	0.0415
IsolationForest normal-only statistical	0.0011	1.0000	0.0022	0.9772	0.0000	0.0917

**Fig. 5.** Single-signal rule/statistical baselines on CT&T test04. They can trigger on anomalies, but their false-alarm behavior is not competitive with the supervised GRAIN representation.

4.8 Feature Ablation and Window Sensitivity

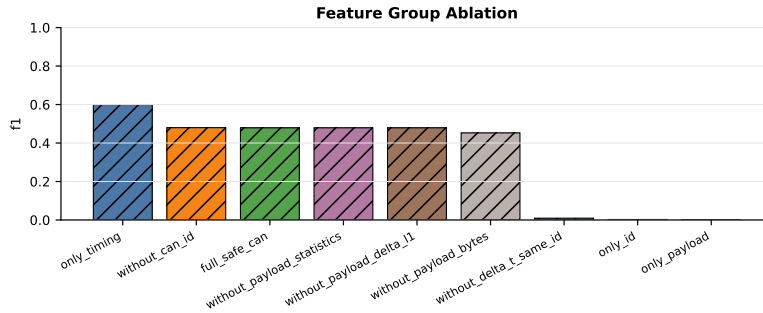
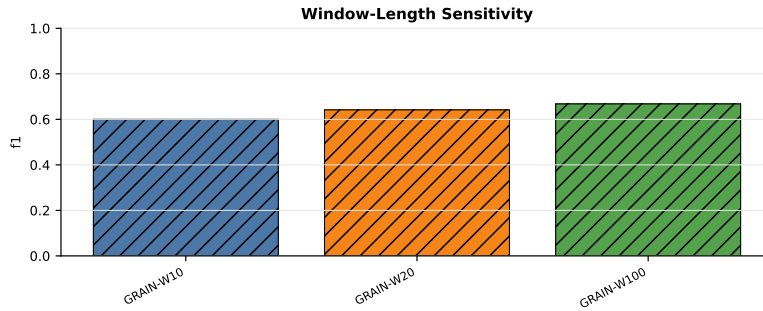
Feature-removal retraining identifies same-ID timing as the strongest sample-level signal in the current CT&T test04 setup. Payload and ID behavior contribute less consistently in these capped-negative runs. Window-length rows are sensitivity analyses, not test-time selection.

Table 10. GRAIN feature-group retraining ablation on CT&T test04.

Variant	P	R	F1	AUPR	R@1e-3
only_timing	0.7708	0.4909	0.5998	0.4246	0.4909
without_can_id	0.7395	0.3550	0.4797	0.2649	0.3550
full_safe_can	0.7403	0.3540	0.4789	0.2547	0.3550
without_payload_statistics	0.7403	0.3540	0.4789	0.2547	0.3550
without_payload_delta_ll	0.7403	0.3540	0.4789	0.2547	0.3550
without_payload_bytes	0.7363	0.3265	0.4524	0.3752	0.4909
without_delta_t_same_id	0.0048	0.2486	0.0094	0.0570	0.0668
only_id	0.0013	0.2555	0.0026	0.0438	0.0550

Table 11. Window-length sensitivity. Rows are not used for test-time window selection.

Setting	Model	P	R	F1	AUPR
ctt_test01	GRAIN-W10	0.9781	0.9533	0.9655	0.9330
ctt_test01	GRAIN-W20	0.9941	0.9442	0.9685	0.9413
ctt_test01	GRAIN-W100	0.9988	0.8919	0.9423	0.8960
ctt_test02	GRAIN-W10	0.9959	0.9937	0.9948	0.9901
ctt_test02	GRAIN-W20	0.9955	0.9899	0.9927	0.9866
ctt_test02	GRAIN-W100	0.9836	0.9306	0.9564	0.9354
ctt_test03	GRAIN-W10	0.9636	0.5212	0.6765	0.6778
ctt_test03	GRAIN-W20	0.9934	0.5028	0.6677	0.7094
ctt_test03	GRAIN-W100	0.9996	0.6623	0.7967	0.9250
ctt_test04	GRAIN-W10	0.8892	0.4529	0.6001	0.6185

**Fig. 6.** Feature-group ablation. Same-ID timing is the dominant sample-level signal in this experiment.**Fig. 7.** Window-length sensitivity on CT&T test04. These rows show sensitivity, not a test-selected hyperparameter search.

4.9 Supplementary Operating Analysis

Score-based AUPR, AUROC, and fixed-FPR detection rates are reported only when scores are available. They are not compared to prior papers unless predictions or scores allow recomputation under the same rule.

Table 12. Supplementary score-based metrics.

Setting	Model	AUPR	AUROC	R@1e-4	R@1e-3
ctt_test01	GradientBoosting / sample	0.9117	0.9897	NA	NA
ctt_test01	GradientBoosting / sample	0.8088	0.8755	NA	NA
ctt_test01	GradientBoosting / window_10	0.9330	0.9765	NA	NA
ctt_test01	GradientBoosting / window_100	0.8960	0.9462	NA	NA
ctt_test01	GradientBoosting / window_20	0.9413	0.9723	NA	NA
ctt_test01	GradientBoosting / window_5	0.9301	0.9825	NA	NA
ctt_test01	GradientBoosting / window_50	0.9207	0.9590	NA	NA
ctt_test01	MLP / sample	0.3565	0.6208	NA	NA

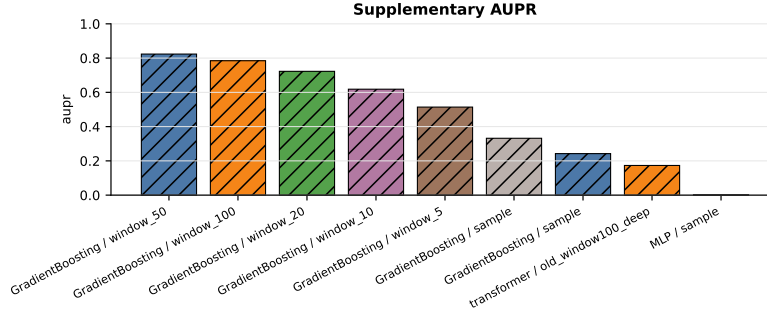


Fig. 8. Supplementary score-based analysis. AUPR and fixed-FPR detection rates are useful deployment evidence, but only when scores are available under the same protocol.

4.10 Metric Availability and Prior Comparability

Not every reproduced row has calibrated scores, and not every prior result exposes the positive class, averaging rule, or confusion matrix. The paper therefore separates conventional attack-positive metrics from supplementary score-based metrics and does not directly compare attack-positive F1 to an externally reported F1 unless the metric definition is recoverable.

Table 13. Metric availability audit.

Setting	Model	attack_positive_F1_computed	AUPR_computed	AUROC_computed	fixed_FPR	detection_rate_computed
ctt_test01	GradientBoosting / sample	True	True	True		False
ctt_test01	GradientBoosting / sample	True	True	True		False
ctt_test01	GradientBoosting / window_10	True	True	True		False
ctt_test01	GradientBoosting / window_100	True	True	True		False
ctt_test01	GradientBoosting / window_20	True	True	True		False
ctt_test01	GradientBoosting / window_5	True	True	True		False
ctt_test01	GradientBoosting / window_50	True	True	True		False
ctt_test01	MLP / sample	True	True	True		False

Table 14. Prior-result comparability audit.

work	dataset_setting	reported_metrics	directly_comparable_to_our_attack_positive_F1
can-train-and-test	Table 13 CT&T test04	accuracy, precision, recall, F1	False

5 Discussion and Limitations

GRAIN-CAN reduces, but does not eliminate, dependence on vehicle-specific absolute CAN distributions. It does not guarantee detection of arbitrary unseen attacks. If an unseen attack preserves timing, payload continuity, and local ID behavior, it may be missed. The method is supervised and uses normal/attack labels; unknown-attack generalization relies on shared local disruptions, not attack-name recognition.

The experimental evidence also has boundaries. First, the feature ablation is a controlled retraining ablation for the available CT&T test04 protocol, but it is not proof that the same feature ordering holds on every vehicle or attack family. Second, score-based rows depend on whether a model exposes comparable scores; missing score metrics are marked unavailable rather than inferred. Third, prior reported F1 values are not directly comparable without positive-class definition, averaging rule, confusion matrix, and predictions or scores. Fourth, event-level deployment evidence still requires official event boundaries for a definitive operational claim. Finally, window size and threshold must be fixed before testing; window sensitivity should not be written as test-time selection.

6 Conclusion

This paper refocuses CAN IDS evaluation around GRAIN-CAN as a supervised feature-based method. The evidence supports same-ID local behavior features before window aggregation as a useful design for cross-shift CAN IDS, while also showing clear limits under unknown-attack test04.

References

1. Alkhatib, N., Mushtaq, M., Ghauch, H., Danger, J.L.: CAN-BERT do it? controller area network intrusion detection system based on BERT language model. arXiv preprint arXiv:2210.09439 (2022)
2. Althunayyan, M., Javed, A., Rana, O.: A robust multi-stage intrusion detection system for in-vehicle network security using hierarchical federated learning. arXiv preprint arXiv:2408.08433 (2024)

3. Checkoway, S., McCoy, D., Kantor, B., Anderson, D., Shacham, H., Savage, S., Koscher, K., Czeskis, A., Roesner, F., Kohno, T.: Comprehensive experimental analyses of automotive attack surfaces. In: *Proceedings of the USENIX Security Symposium* (2011)
4. Cho, K.T., Shin, K.G.: Fingerprinting electronic control units for vehicle intrusion detection. In: *Proceedings of the USENIX Security Symposium*. pp. 911–927 (2016)
5. Cho, K.T., Shin, K.G.: Viden: Attacker identification on in-vehicle networks. In: *Proceedings of the ACM Conference on Computer and Communications Security*. pp. 1109–1123 (2017)
6. Davis, J., Goadrich, M.: The relationship between precision-recall and ROC curves. In: *Proceedings of the International Conference on Machine Learning*. pp. 233–240 (2006)
7. Guerra, L., Xu, L., Bellavista, P., Chapuis, T., Duc, G., Mozharovskiy, P., Nguyen, V.T.: AI-driven intrusion detection systems on the ROAD dataset: A comparative analysis for automotive CAN. *arXiv preprint arXiv:2408.17235* (2024)
8. Hossain, N., Moriano, P.: Evaluating false alarm and missing attacks in CAN IDS. *arXiv preprint arXiv:2602.02781* (2026)
9. Kidmose, E., et al.: can-sleuth: Benchmarking CAN intrusion detection. Public project and benchmark documentation (2025)
10. Koscher, K., Czeskis, A., Roesner, F., Patel, S., Kohno, T., Checkoway, S., McCoy, D., Kantor, B., Anderson, D., Shacham, H., Savage, S.: Experimental security analysis of a modern automobile. In: *Proceedings of the IEEE Symposium on Security and Privacy*. pp. 447–462 (2010)
11. Lampe, B., Meng, W.: can-train-and-test: A curated CAN dataset for automotive intrusion detection. *Computers & Security* **140**, 103777 (2024)
12. Liu, Q., Song, R., Cui, L., Zhang, H., Sun, Y., Sun, L.: MIDS: Detecting stealthy masquerade and tampering attacks on CAN bus via bidirectional Mamba. *arXiv preprint arXiv:2606.18599* (2026)
13. Miller, C., Valasek, C.: Remote exploitation of an unaltered passenger vehicle. In: *Black Hat USA* (2015)
14. Saito, T., Rehmsmeier, M.: The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE* **10**(3), e0118432 (2015)
15. Shahriar, M.H., Xiao, Y., Moriano, P., Lou, W., Hou, Y.T.: CANShield: Deep learning-based intrusion detection framework for controller area networks at the signal-level. *arXiv preprint arXiv:2205.01306* (2022)
16. Sharmin, S., Mansor, H., Abdul Kadir, A.F., Aziz, N.A.: Benchmarking frameworks and comparative studies of controller area network intrusion detection systems: A review. *arXiv preprint arXiv:2402.06904* (2024)
17. Song, H.M., Kim, H.R., Kim, H.K.: Intrusion detection system based on the analysis of time intervals of CAN messages for in-vehicle network. In: *International Conference on Information Networking*. pp. 63–68 (2016)
18. Taylor, A., Leblanc, S., Japkowicz, N.: Frequency-based anomaly detection for the automotive CAN bus. In: *World Congress on Industrial Control Systems Security*. pp. 45–49 (2015)
19. Verma, M.E., Bridges, R.A., Hollifield, S.C., Iannacone, M.D., Moriano, P.: ROAD: The real ORNL automotive dynamometer controller area network intrusion detection dataset. *arXiv preprint arXiv:2012.14600* (2020)
20. Wang, K., Jiang, Q., Wang, B., Zhang, Y., Wu, Y.: Effective in-vehicle intrusion detection via multi-view statistical graph learning on CAN messages. *arXiv preprint arXiv:2311.07056* (2023)